

Deep Learning Workshop: Project Proposal

Early-Epoch Tidal Disruption Event Classification

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1 Introduction

The field of Astronomy is expecting a massive data boom with the construction of the Vera C. Rubin Observatory. The telescope’s purpose is to create a 10 year time-lapse record of the southern sky, gathering 10 terabytes of data every day. This raises the need for an efficient classification solution for irregular time-series data streams.

For this end, the Astronomical community has hosted several open sources data challenges to jump start development. The Photometric LSST Astronomical Time-Series Classification Challenge (PLAsTiCC) established the benchmark for classifying millions of simulated light curves. Its successor, ELAsTiCC, introduced real-time streaming constraints. Recently, the MALLORN dataset was released, focusing on imbalanced datasets of rare astronomical events.

This project aims to leverage these open datasets to build an early-epoch deep learning classifier capable of identifying specific astronomical transients dynamically as new observations arrive. Specifically, identifying Tidal Disruption Events (TDE) at their beginning, before they fade away.

2 Astrophysical background

Telescopes track celestial events by measuring brightness (flux) in specific wavelengths, taken at irregular intervals (triggered by flux changes). Relevant terms include:

- **Supernovae:** Exploding stars that exhibit a well understood rise and fall in brightness over weeks or months. These are common in the datasets.
- **Active Galactic Nuclei (AGNs):** Supermassive black holes actively consume surrounding gas, creating a chaotic, flickering light curve that lasts for years. These act as the primary “background noise.”

- **Tidal Disruption Events (TDEs):** An extremely rare anomaly in which a star wanders too close to a supermassive black hole and is shredded. This creates a distinct, long-lasting thermal flare.

3 End Product

The primary objective is to build an Early Epoch DL sequence classifier, in order to identify TDEs before their peak luminosity. If high confidence is achieved on the simulated data, the pipeline can be extended to validate its real-world performance against newly available, noisy stream data from the actively operating Vera C. Rubin Observatory (LSST).

Anticipating unforeseen challenges throughout the project, we also suggest possible alternative directions. These potential fallbacks include unsupervised anomaly detection for out-of-distribution event tracking, full light curve analysis utilizing complete sequential profiles, or standard multi-class transient classification utilizing the generalized PLAsTiCC benchmark.

4 Training and Inference Schemes

Dealing with time sequence data, the methodology will rely on Recurrent Neural Networks (RNNs) and Transfer Learning.

Because the time intervals between telescope observations vary, the model cannot assume fixed time steps. The network will ingest an input vector at each step comprising the time difference (Δt), flux, flux error, and the passband filter.

Methodology:

- **Pre-training:** The network will first be trained on the massive, generalized PLAsTiCC dataset. This allows the foundational layers to learn the “language” of light curves, internalizing how to process noisy flux measurements and irregular gaps.
- **Fine-tuning:** To address the severe class imbalance of hunting for rare anomalies, the pre-trained model will be fine-tuned on the much smaller MALLORN dataset. The underlying temporal processing layers will be frozen, and the final classification layers will be optimized to isolate TDEs from the AGN background.

5 Datasets

The project will utilize the following open-source datasets:

- **PLAsTiCC (2018):** Contains approximately 3.5 million simulated light curves across over a dozen transient classes. It will serve for the pre-training stage.

- **MALLORN (2025):** Contains roughly 10,000 simulated light curves based on real observations from the Zwicky Transient Facility (ZTF). The dataset is highly imbalanced, with rare Tidal Disruption Events (TDEs) among more common transient types.
- **Vera C. Rubin Observatory Observations (2026):** Real life observations.

6 Compute and Storage

The entire pipeline will be developed locally, with the following specs:

- **GPU:** NVIDIA RTX 5070 Ti
- **CPU:** AMD Ryzen 7 7800X3D
- **Memory:** 32 GB RAM
- **Storage:** 500 GB of available local SSD storage

7 Third Party Tools and Models

- **Deep Learning Framework:** PyTorch (for constructing and training the RNNs/Transformers). GitHub and PyPI to pull baseline models.
- **Data Processing:** Pandas, NumPy, and Scikit-Learn for data wrangling, normalization, and handling class imbalances.
- **Astrophysics Libraries:** Astropy (for standardized handling of astronomical time formats and tables).

8 Existing Solutions

Notable Machine Learning Models (Post-PLAsTiCC):

- **RAPID (Recurrent Autoencoder for Photometric ID):** A model geared toward early classification. It utilizes Gated Recurrent Units (GRUs) to ingest the light curve sequentially, outputting an updated classification probability at every time step.
- **tdescore:** A recent, highly specialized classifier designed specifically for the needle-in-a-haystack problem of finding Tidal Disruption Events (TDEs). It focuses heavily on anomaly detection within nuclear transient streams.
- **Avocado:** The winning model of the original PLAsTiCC challenge. It uses traditional approach with Gaussian Processes and classifying the extracted features using LightGBM. Works with full-curve classification.

9 Literature

Given the advanced nature of these papers, understanding the theoretical aspects is beyond the scope of this project. This literature will serve for specific architectural guidance when encountering roadblocks, and be used as an academic reference to compare our results.

9.1 Key Papers (TDEs & Early Classification)

- *tdescore: An Accurate Photometric Classifier for Tidal Disruption Events* (Stein et al., 2024)
<https://iopscience.iop.org/article/10.3847/2041-8213/ad3337>
- *RAPID: Early Classification of Explosive Transients using Deep Learning* (Muthukrishna et al., 2019)
<https://arxiv.org/abs/1904.00014>
Pioneers the use of Recurrent Neural Networks (GRUs) to provide dynamic, early-epoch time-series classifications as new data arrives.
- *The automation of optical transient discovery and classification in Rubin-era time-domain astronomy* (Rehemtulla et al., 2025)
<https://www.nature.com/articles/s41550-025-02720-6>

9.2 Foundational Datasets (PLAsTiCC & ELAsTiCC)

- *The PLAsTiCC Data Set* (The PLAsTiCC Team, 2018)
<https://arxiv.org/abs/1810.00001>
- *Models and Simulations for the PLAsTiCC Challenge* (Kessler et al., 2019)
<https://arxiv.org/abs/1903.11756>
- *ELAsTiCC Data Portal* (LSST DESC)
https://portal.nersc.gov/cfs/lsst/DESC_TD_PUBLIC/ELASTICC/